



AI THREAT INTELLIGENCE & RED TEAM PLATFORM

MEDFLOW INTERNSHIP –
COMPLETE SETUP & BUILD
GUIDE





PLAN

- What We Build in This Module
- System Requirements
- Environment Setup & Install Python Packages
- Download MITRE CTI Data
- The 4 Vector Databases
- Red Team Agents & Threat Intelligence Agents
- Healthcare Test Scenarios

What We Build?

Red Team Side	Threat Intelligence Side
• Adversary simulation agent using MITRE ATT&CK • Understands sub-techniques, tools, kill chains • how does an attacker do X step by step? • Powered by redteam Database	• CTI analyst agent for technique lookup • Attribution agent: who is behind the attack • Detection agent: how to detect and respond

SYSTEM REQUIREMENTS

Requirement	Detail
Python	3.10 or higher
RAM	16 GB recommended
Groq Account	Free at console.groq.com – gives API key
Git	Required to clone MITRE CTI repository



ENVIRONMENT SETUP & INSTALL PYTHON PACKAGES

Package

Chromadb: recommended

sentence-transformers

groq

python-dotenv

Purpose

Vector database – stores and searches embeddings

Converts text to vectors using
BAAI/bge-base-en-v1.5 : recommended

Python client for the Groq LLM API

Reads your API key from .env file securely

DOWNLOAD MITRE CTI DATA

official repository:
<https://github.com/mitre/cti>

MITRE | ATT&CK[®]

THE 4 VECTOR DATABASES

Database	What it indexes
attack_db	858 techniques
redteam_db	10 000+ sub-techniques + procedures
actor_db	189 actors + 729 malware + 95 tools
detection_db	1758 analytics + 699 strategies + 268 mitigations

RED TEAM AGENT

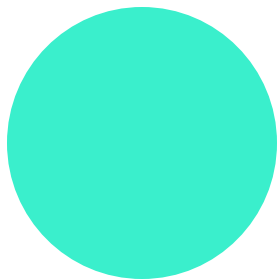
Operational red team knowledge – goes beyond definitions into how attacks are actually executed

- Sub-technique lookup
- Tool mapping: which techniques Mimikatz, Cobalt Strike, TrickBot implement
- Kill chain generation: full attack chain
- Procedure examples: real-world steps extracted

How to validate: Ask questions!!



THREAT INTELLIGENCE AGENTS



- Technique definition, tactic mapping, detection guidance, mitigation overview, healthcare context framing
- Ranked attribution scoring, TTP overlap matching, tool fingerprinting, campaign correlation, healthcare sector filtering
- SIEM query guidance, data source recommendations, detection logic per technique, healthcare-specific IOCs, mitigation mapping, response action recommendations

How to **validate**: Ask questions!!

HEALTHCARE TEST SCENARIOS

Red team questions:

e.g. Give me a full kill chain for a Ryuk attack on a hospital.

Threat intelligence questions:

e.g. What SIEM rules detect MFA fatigue attacks against hospital portals

Hallucination tests :

e.g. what is T9999